

Welcome to Code Crunch!

C[6] | CYBER CRUNCH: Networking Fundamentals

[Thu 1-2PM] Weekly | Spring 2025

Attendance



linktr.ee/CODE.CRUNCH

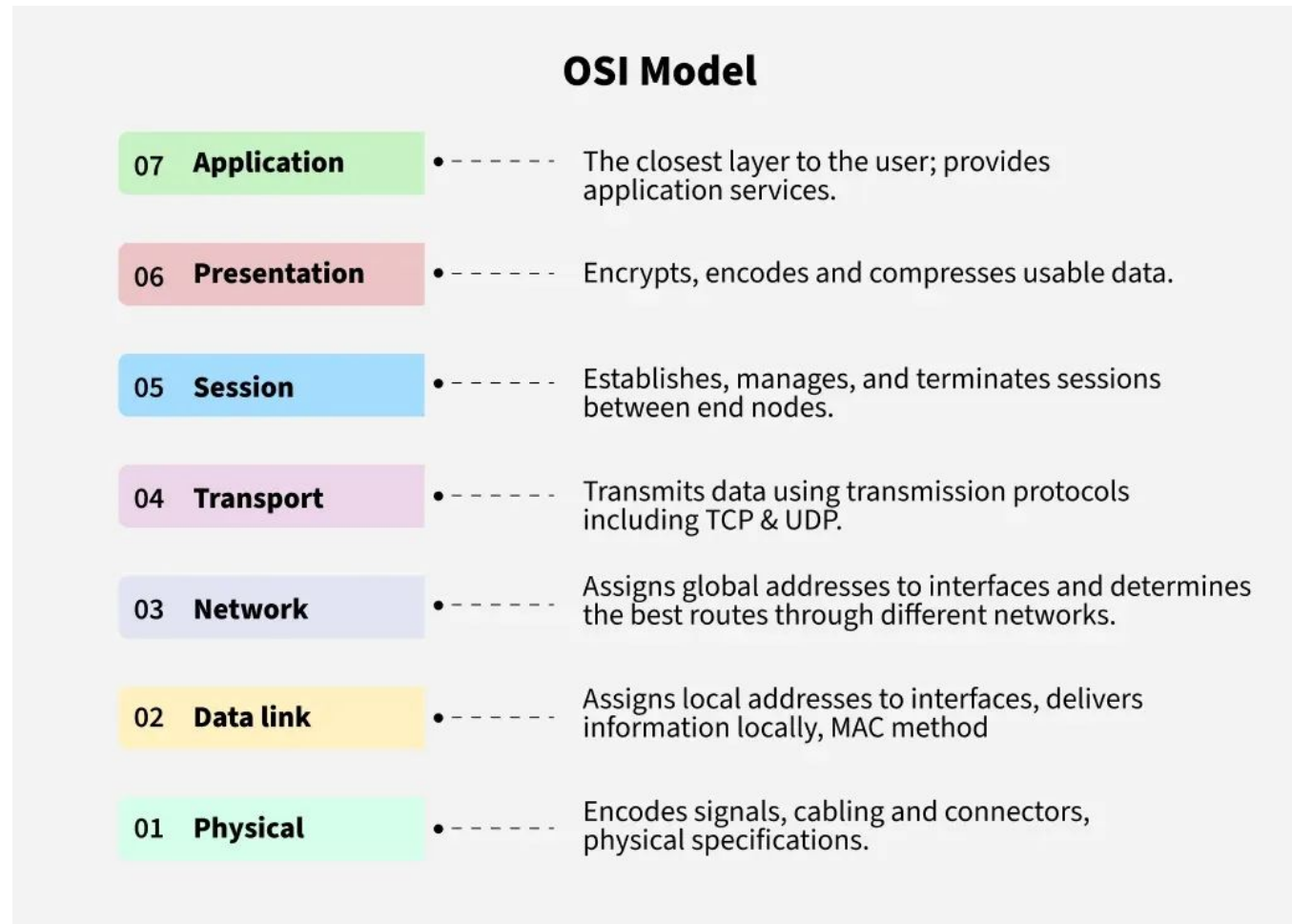


**CODE
CRUNCH**

OSI Model



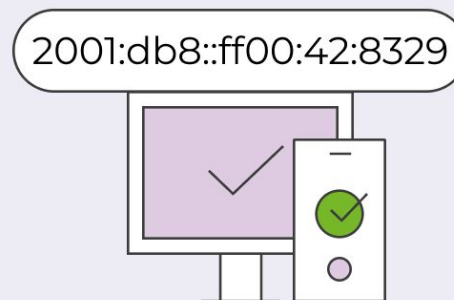
- The **OSI (Open Systems Interconnection)** Model is a set of rules that explains how different computer systems communicate over a network.



What is Networking?

- **Definition:** Connecting devices to share resources and communicate.
- **Key Concepts:**
 - IP Addressing (IPv4/IPv6).

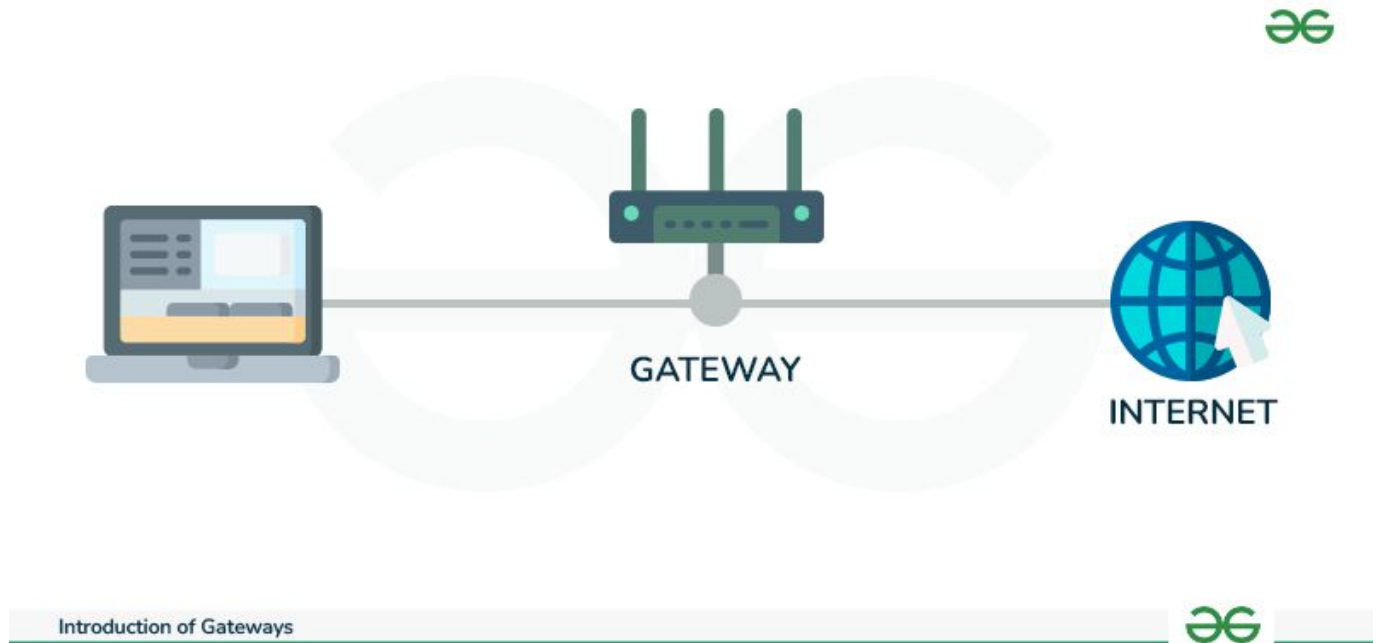
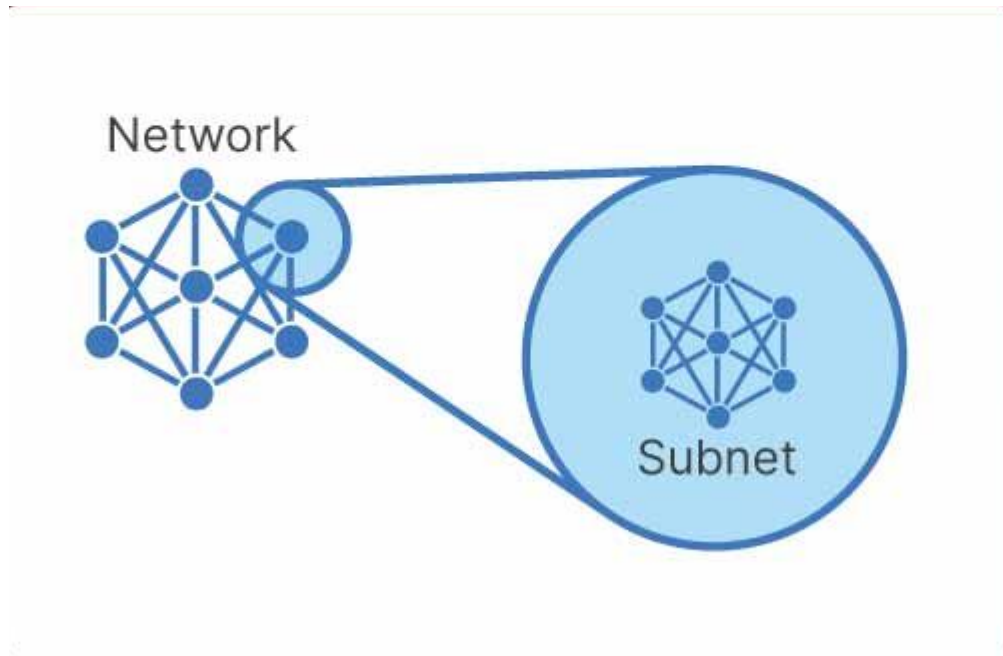
IPV4 VS IPV6



Key Concepts

Subnets:

Gateways:

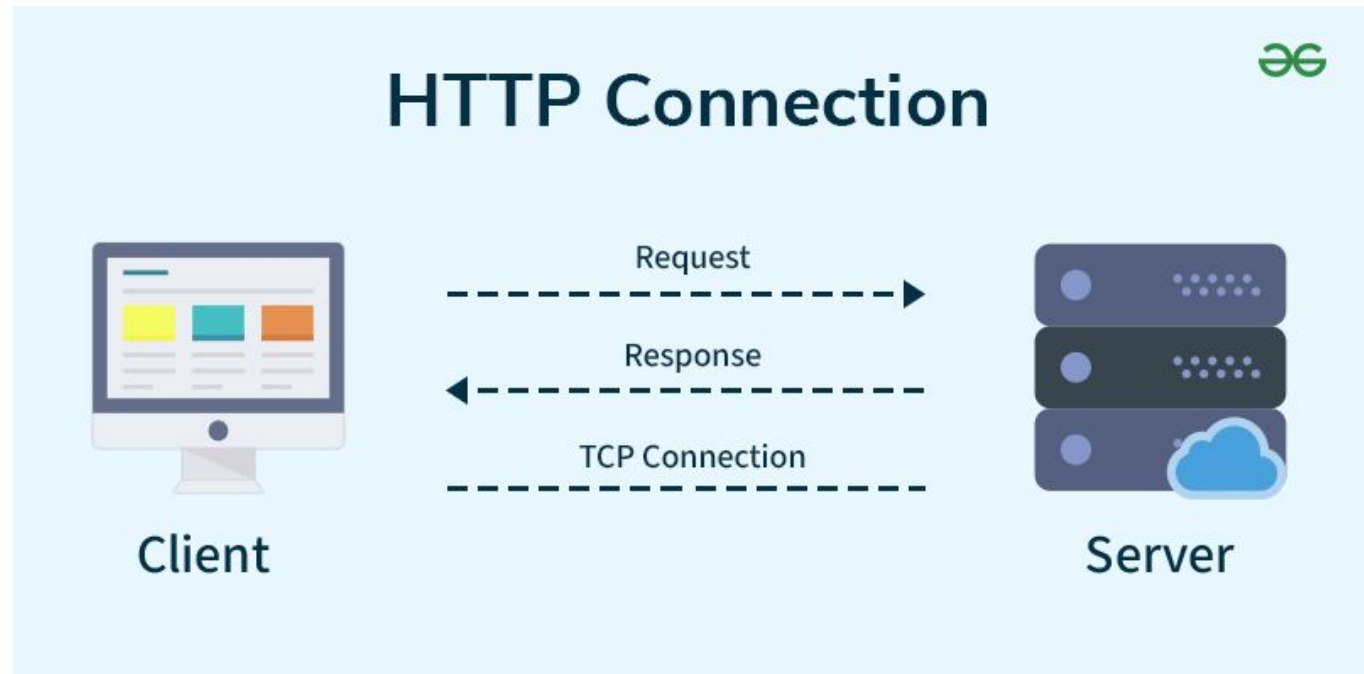


Key Concepts



- **Protocols:**

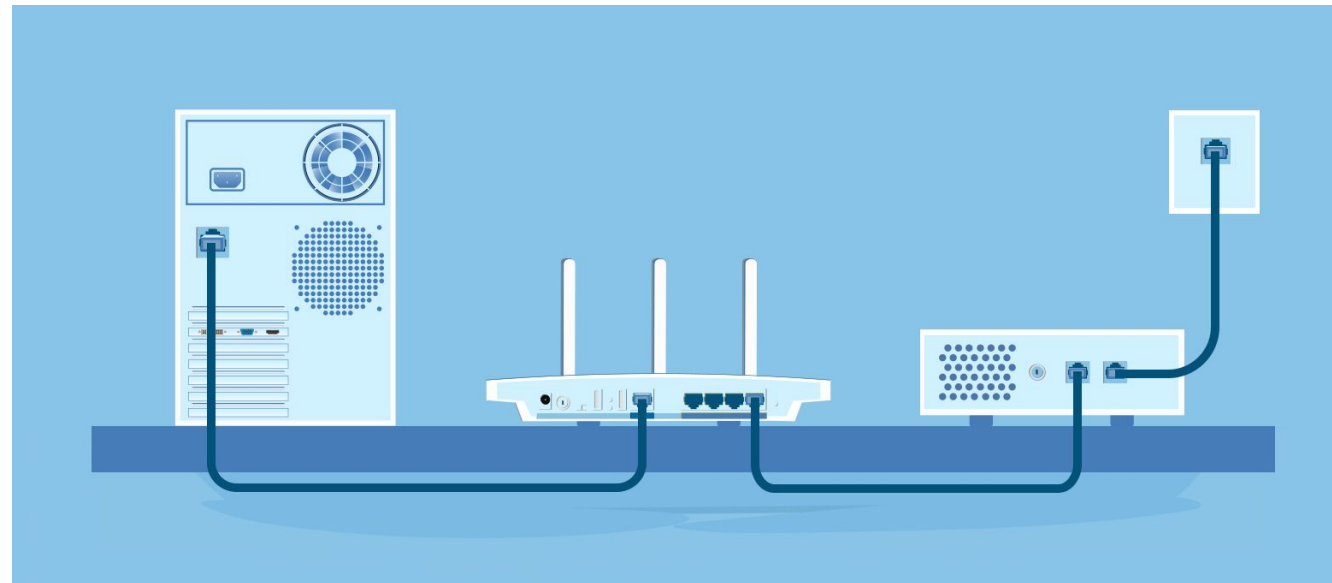
- **TCP/IP:** Transmission Control Protocol/ Internet Protocol
- **HTTP:** Hypertext Transfer Protocol
- **HTTPS:** Hypertext Transfer Protocol Secure
- **DNS:** Domain Name System



Importance of networking in Cybersecurity



- Monitoring traffic to identify anomalies.
- Understanding protocols to secure communications.
- Identifying vulnerabilities in network configurations.





Hands-On Activity Overview

Platform: TryHackMe

<https://tryhackme.com/r/room/introtonetworking>

- **Room:** "Introduction to Networking"
- **Concepts Covered:** IP addressing, common protocols, and network communication basics.
- **Steps:**
 1. Follow instructor guidance through initial sections.
 2. Spend 15-20 minutes independently completing tasks.
 3. Share insights and discuss as a group.

Visual: Screenshot of the TryHackMe Networking room.



Debrief and Key Takeaways

- **Key Learnings:**

- Networking is foundational for cybersecurity.
- Tools like Wireshark are essential for traffic analysis and incident response.

- **Next Steps:**

- Explore deeper networking topics on TryHackMe or through additional practice.
- Practice packet analysis with real or simulated data.

Real-World Application: Monitoring and securing enterprise networks.

Visual: Checklist or summary graphic.



Q&A Session



Attendance



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Join us for the next sessions



Thursdays: 11AM – 12PM & 2PM – 3PM

Friday: 3PM – 4PM

Monday: 12PM – 1PM



RSVP lu.ma/CODECRUNCH



Your feedback helps us improve. Share your thoughts!

Go to our website: go.fiu.edu/codecrunch